

ELEC4631 Lecture 6: Lyapunov Stability Analysis

Part 2

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Outline

- Extension of Lyapunov stability analysis
- Krasovskii-LaSalle invariance principle

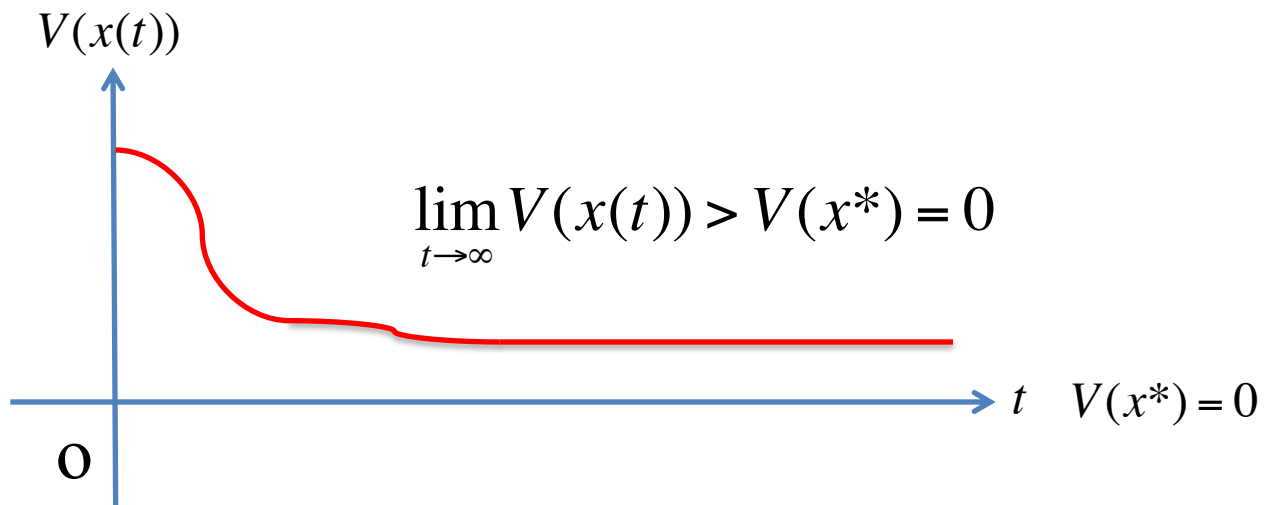
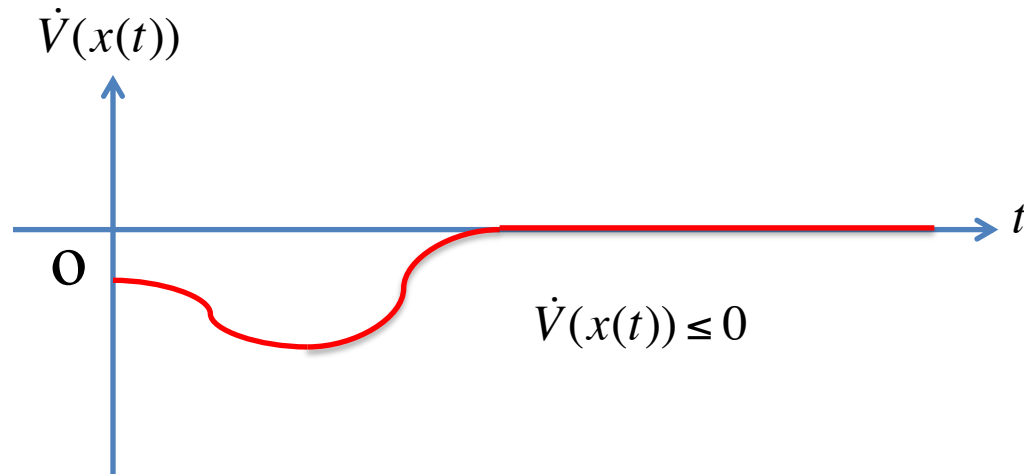
Limitations of fundamental theorem of Lyapunov

- Constructing a Lyapunov function satisfying $\dot{V}$ is lnd or nd around x^* to prove asymptotic or global asymptotic stability is in general challenging, even with physically motivated candidate functions
- It is often easier to find a candidate Lyapunov V with $\dot{V}$ lnsd or nsd. Is it still possible to establish asymptotic or global asymptotic stability in this case?

Possible refinement of theorem

- If $\dot{V}(x(t)) \leq 0$ theorem only guarantees that system is stable i.s.L around x^* .
- However, often with more analysis it may be possible to go further to show that system is asymptotically stable. Principle will be illustrated via some examples.

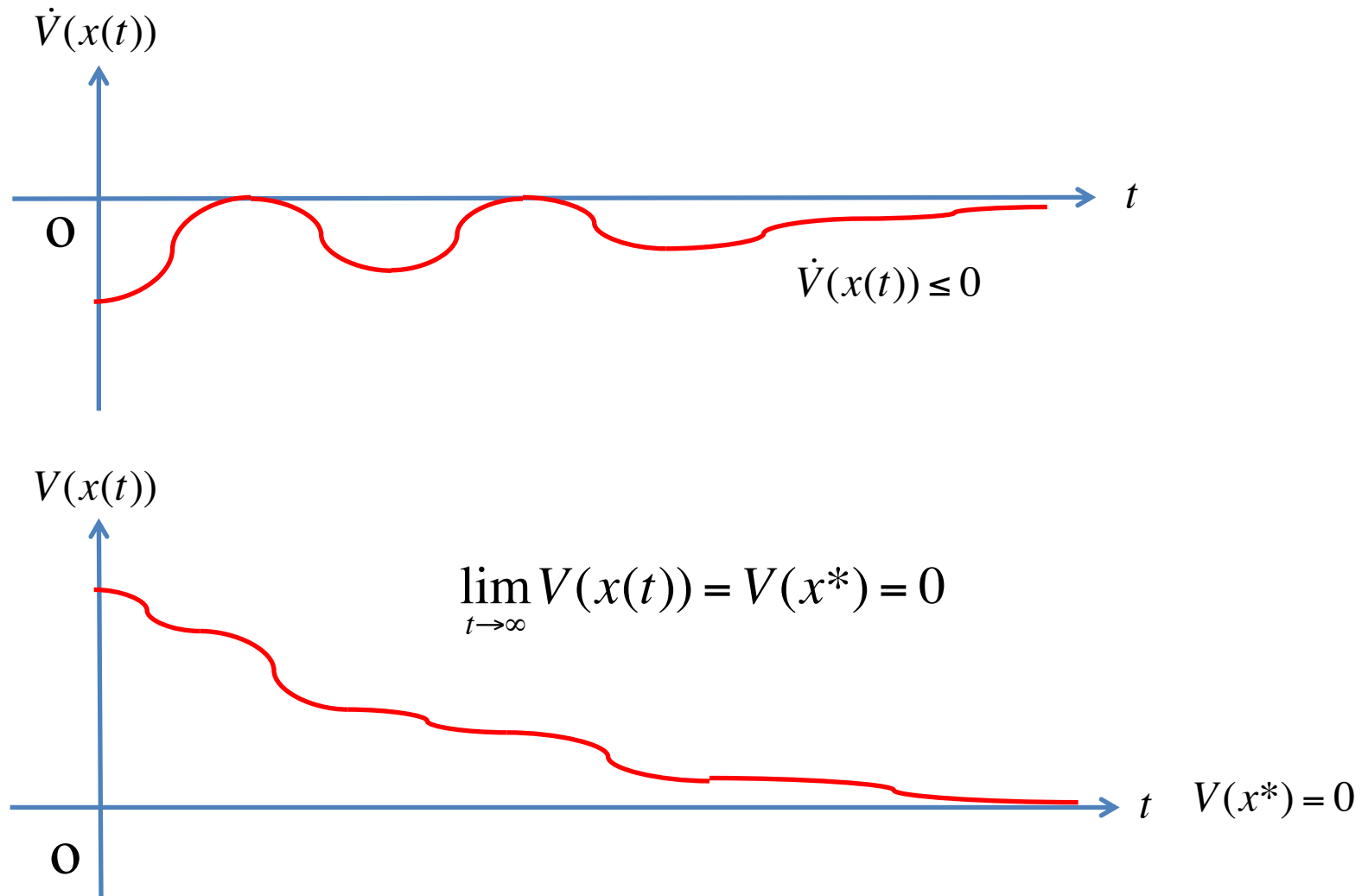
Derivative of $V(x(t))$ lnsd, $x(t)$ can fail to converge to x^* as $t \rightarrow \infty$



Possible refinement of theorem

- Main idea: if $\dot{V}(x(t)) = 0$ for $x(t) \neq x^*$ only occurs for an instant before immediately switching to a negative value $\dot{V}(x(t^+)) < 0$ then $V(x(t))$ continues to decrease over time until reaching $V(x^*) = 0$.
- Principle above can be formulated precisely and known as *Krasovskii-LaSalle Invariance Principle*.

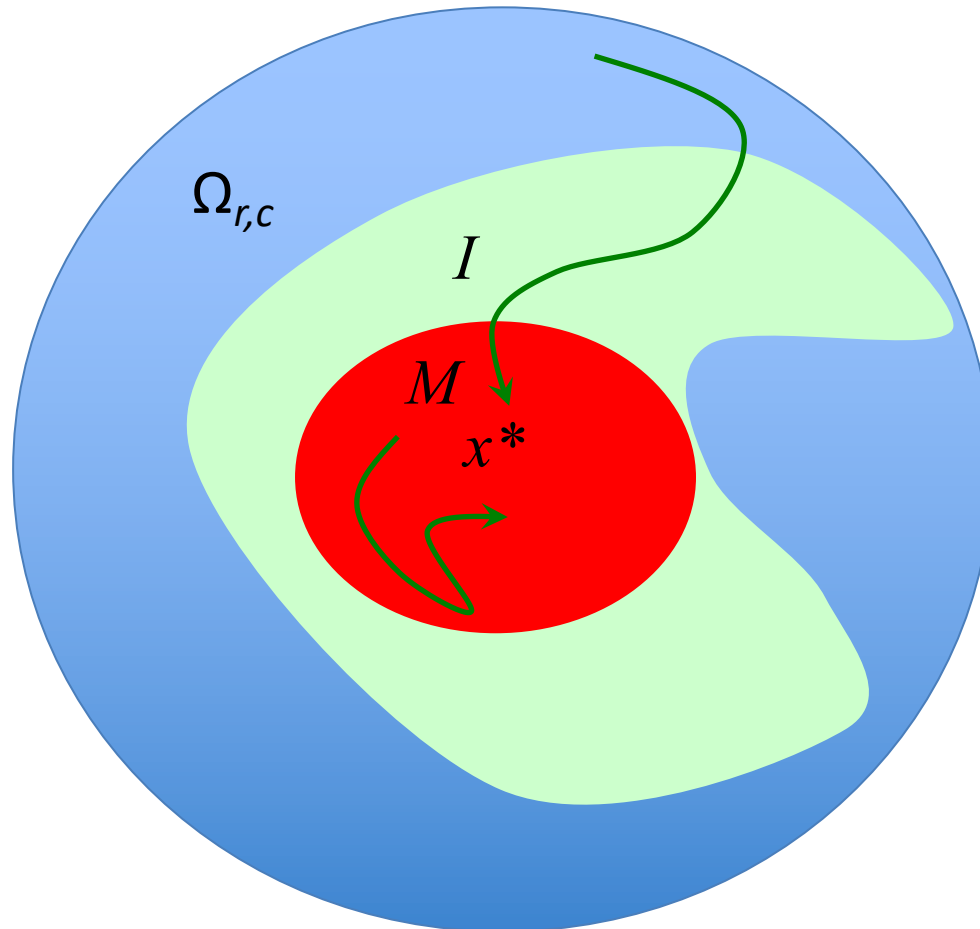
Derivative of $V(x(t))$ is not zero but $x(t)$ still converges to x^* as $t \rightarrow \infty$



Krasovskii-LaSalle Invariance Principle

- If $V(x)$ is lpd and $\dot{V}(x) = \nabla V(x)^T f(x)$ is lnsd in a set $\Omega_{r,c} = \left\{ x \in R^n \mid \|x - x^*\| \leq r, V(x) \leq c \right\} \subseteq R^n$ ($r, c > 0$) containing an equilibrium point x^* , and $\Omega_{r,c}$ is positively invariant, then trajectories starting in $\Omega_{r,c}$ converge to the largest positively invariant set M in the set $I = \left\{ x \in \Omega_{r,c} \mid \dot{V}(x) = 0 \right\}$.
- A set C being positively invariant means that if $x(0)$ is in C then $x(t)$ is in C for all $t \geq 0$.

Krasovskii-LaSalle Invariance Principle



Krasovskii-LaSalle Invariance Principle

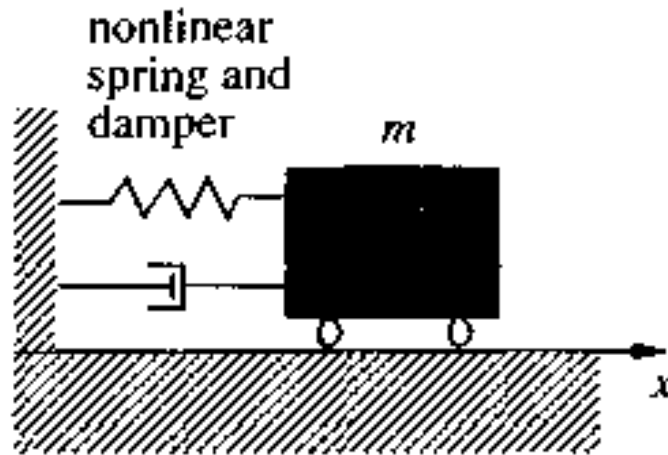
- M is a set that contains all points that lie on trajectories satisfying $x(t) \in I \quad \forall t \geq 0$
- To find M , typically one needs to first *find all such trajectories*. Then any point that is passed by these trajectories at any time $t \geq 0$ is an element of M .

Krasovskii-LaSalle Invariance Principle

- If the only trajectory in M is the trajectory $x(t) = x^* \forall t \geq 0$ then $M = \{x^*\}$ and x^* is *asymptotically stable*. If in addition $V(x)$ is a global Lyapunov function around x^* and radially unbounded then x^* is *globally asymptotically stable*.

Example 1

- Consider a mass-spring-damper system below



with a nonlinear spring and damper.

- It is described by the nonlinear ODE

$$m\ddot{x} + b\dot{x}|\dot{x}| + k_0x + k_1x^3 = 0$$

Example 1

- Introducing the state

$$\xi = (\xi_1, \xi_2)^T, \quad \xi_1 = x, \xi_2 = \dot{x}$$

the system can be written in the state-space form

$$\dot{\xi} = \begin{bmatrix} \xi_2 \\ -\frac{b}{m}\xi_2|\xi_2| - \frac{k_0}{m}\xi_1 - \frac{k_1}{m}\xi_1^3 \end{bmatrix}$$

Example 1

- Consider the candidate Lyapunov function

$$V(\xi) = \frac{1}{2}m\xi_2^2 + \frac{1}{2}k_0\xi_1^2 + \frac{1}{4}k_1\xi_1^4$$

- This candidate is pd
- Compute $\dot{V}$

$$\dot{V}(\xi) = m\xi_2\dot{\xi}_2 + k_0\xi_1\dot{\xi}_1 + k_1\xi_1^3\dot{\xi}_1 = -b|\xi_2|^3$$

Example 1

- We have

$$\dot{V}(\xi) = 0 \Leftrightarrow \xi_2 = 0$$

and $\dot{V}$ is nsd (not nd)

- Consider a set containing $\xi = (0,0)^T$,

$$\Omega_{r,c} = \{\xi \in \mathbb{R}^2 \mid \|\xi\| \leq r, V(\xi) \leq c\}$$

for some $r, c > 0$. We seek to describe the largest positively invariant set M contained in

$$I = \{\xi \in \Omega_{r,c} \mid \dot{V}(\xi) = 0\}$$

Example 1

- A trajectory that starts in I must have $\xi_2(0) = 0$. To stay in $I \forall t > 0$ we must have $\xi_2(t) = 0$ for $\forall t > 0$. This implies that $\dot{\xi}_2(t) = 0 \forall t \geq 0$
- Substituting $\xi_2(t) = 0$ and $\dot{\xi}_2(t) = 0 \forall t \geq 0$ into the state-space equation gives

$$\begin{bmatrix} \dot{\xi}_1(t) \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{k_0}{m} \xi_1(t) - \frac{k_1}{m} \xi_1(t)^3 \end{bmatrix}$$

Example 1

- We conclude that $\xi_1(t) = c \ \forall t \geq 0$ for some constant c satisfying $-(k_0/m) c - (k_1/m) c^3 = 0$. This gives $c = 0$, thus $\xi_1(t) = 0 \ \forall t \geq 0$.
- The only trajectory that is positively invariant inside I is the constant trajectory $\xi(t) = (0,0)^T \ \forall t \geq 0$. This means that M contains only a single point: $M = \{(0,0)^T\}$.
- By the K-L invariance principle, the origin is asymptotically stable

Example 1

- In fact for this example we can go even further.
- The argument holds for any choice of $r, c > 0$ and since $V(\xi)$ is radially unbounded ($V(\xi) \rightarrow \infty$ as $\|\xi\| \rightarrow \infty$), we can choose r and c as large as desired
- Therefore, by the K-L invariance principle the origin is not only asymptotically stable, it is globally asymptotically stable!

Example 2: Stability of RLC circuit

- Recall the RLC circuit ODE:

$$\begin{aligned}\dot{v}_C &= \frac{1}{C}i, \\ \dot{i} &= -\frac{1}{L}v_C - \frac{R}{L}i,\end{aligned}$$

- The only equilibrium point is $(0,0)$.
- Take the circuit's total stored energy as candidate Lyapunov function defined for all (v_C, i) :

$$\begin{aligned}V(v_C, i) &= \text{energy of capacitor} + \text{energy of inductor}, \\ &= \frac{1}{2}Cv_C^2 + \frac{1}{2}Li^2\end{aligned}$$

Example 2: Stability of RLC circuit

- We then have that for all (v_C, i)

$$\begin{aligned}\dot{V}(x) &= \nabla V(x)^T f(x), \\ &= \begin{bmatrix} \frac{\partial V}{\partial v_C}(x) & \frac{\partial V}{\partial i}(x) \end{bmatrix} \begin{bmatrix} -\frac{1}{L}v_C - \frac{1}{C}i \\ \frac{R}{L}i \end{bmatrix}, \\ &= \begin{bmatrix} Cv_C & Li \end{bmatrix} \begin{bmatrix} -\frac{1}{L}v_C - \frac{1}{C}i \\ \frac{R}{L}i \end{bmatrix}, \\ &= -i^2 R, \\ &\leq 0.\end{aligned}$$

$\dot{V}(x)$ is lnsd but not lnd!

- Thus V is global Lyapunov function for the RLC circuit. Notice that $-i^2 R$ is simply the expression for the power dissipated by the resistor as heat.

Example 2: Stability of RLC circuit

- Since $\dot{V}(x)$ is lnsd, we conclude that the circuit is stable i.s.L around the origin.
- But we cannot yet conclude that it is asymptotically stable as expected from the circuit physics. We can establish that this is the case by a bit more analysis.

Example 2: Stability of RLC circuit

- Now, notice that $\dot{V}(x) = 0 \Rightarrow i = 0$
- Consider the set

$$\Omega_{r,c} = \{ \|x\| \leq r \mid V(x) \leq c \}$$

for an appropriate choice of $r, c > 0$ so that it is positively invariant. Let

$$I = \{ x \in \Omega_{r,c} \mid i = 0 \}$$

- Which trajectories that start in I would remain in I ? We must have that if $i(0) = 0$ then $\dot{i}(t) = 0$ and $i(t) = 0$ for $t \geq 0$. Therefore, $v_C(t) = 0$ for $t \geq 0$. Hence, $M = \{(0,0)\}$ and the origin is asymptotically stable. Moreover, since V is a global Lyapunov function around $(0,0)$ and is radially unbounded $(0,0)$ is in fact globally asymptotically stable.

Example 2: Stability of RLC circuit

- It is possible to consider a Lyapunov function that is non-physically motivated.
- For instance, one could consider a candidate Lyapunov function of the real quadratic form

$$V(v_C, i) = \frac{1}{2} v_C^2 + \varepsilon L v_C i + \frac{L}{2C} i^2$$

for some suitable real constant $\varepsilon > 0$.

Example 2: Stability of RLC circuit

- There is candidate Lyapunov function of this form such that V and $\dot{V}$ are positive and negative definite quadratic forms, respectively, for $\varepsilon > 0$ small enough (nice tutorial exercise).
- By this approach we can again establish that the RLC circuit is globally asymptotically stable around the origin.

Example 3: Stability of pendulum

- Recall the pendulum ODE:

$$\dot{\theta}(t) = \omega(t), \quad \theta(0) = \theta_0$$

$$\dot{\omega}(t) = -\frac{g}{r} \sin(\theta(t)), \quad \omega(0) = \omega_0,$$

- Equilibrium points are $(\theta, \omega) = (2n\pi, 0)$ and $((2n+1)\pi, 0)$ for any integer n , corresponding to the pendulum being in the vertical downward or vertical upward position, respectively.
- We consider equilibrium point $(0, 0)$.

Example 3: Stability of pendulum

- Physics gives us pendulum's total energy as candidate Lyapunov function:

$$\begin{aligned} V(\theta, \omega) &= \text{kinetic energy} + \text{potential energy} \\ &= \frac{1}{2}mr^2\omega^2 + mgr(1 - \cos \theta) \end{aligned}$$

Example 3: Stability of pendulum

- Let $x = (\theta, \omega)$. Then

$$\begin{aligned}\dot{V}(x) &= \nabla V(x)^T f(x), \\ &= \begin{bmatrix} \frac{\partial V}{\partial \theta}(x) & \frac{\partial V}{\partial \omega}(x) \end{bmatrix} \begin{bmatrix} \omega \\ -\frac{g}{r} \sin \theta \end{bmatrix} \\ &= \begin{bmatrix} mgr \sin \theta & mr^2 \omega \end{bmatrix} \begin{bmatrix} \omega \\ -\frac{g}{r} \sin \theta \end{bmatrix} \\ &= 0.\end{aligned}$$

$\dot{V}(x)$ is lnsd but not lnd!

- Thus V is a Lyapunov function.

Example 3: Stability of pendulum

- So, according to theorem pendulum is stable i.s.L around $(0,0)$.
- Not surprising. Since there is no friction -- no energy lost -- physics tells us that energy in system remains constant. So, if pendulum given an initial kick it will oscillate forever.

Example 3: Stability of pendulum

- Now, consider slightly different scenario. Allow a damping torque proportional to velocity (e.g., due to drag by the surrounding air).

$$\dot{\theta}(t) = \omega(t), \quad \theta(0) = \theta_0$$

$$\dot{\omega}(t) = -\frac{g}{r} \sin(\theta(t)) - D\omega(t), \quad \omega(0) = \omega_0,$$

- D is positive damping constant.

Example 3: Stability of pendulum

- Now, $f(\theta, \omega) = \begin{bmatrix} \omega \\ -\frac{g}{r} \sin \theta - D\omega \end{bmatrix}$

and by similar calculations as before

$$\begin{aligned} \dot{V}(x) &= \nabla V(x)^T f(x), \\ &= \begin{bmatrix} mgr \sin \theta & mr^2 \omega \end{bmatrix} \begin{bmatrix} \omega \\ -\frac{g}{r} \sin \theta - D\omega \end{bmatrix}, \\ &= -mr^2 D\omega^2, \\ &\leq 0. \end{aligned}$$

Again, $\dot{V}(x)$ is lnsd but not lnd!

Example 3: Stability of pendulum

- So, still we can only conclude that system is stable i.s.L around the origin.
- However, with drag we expect origin to be asymptotically stable since it dissipates system's energy.
- It turns out that is in fact the case, but as in RLC example this needs a bit more analysis.

Example 3: Stability of pendulum

- Note that $\dot{V}(x) = 0 \Rightarrow \omega = 0$. Take the set $\Omega_{r,c} = \{\|x\| \leq r \mid V(x) \leq c\}$ containing the origin, for some radius $0 < r < \pi$ (so that $\Omega_{r,c}$ does not contain the other equilibrium point) and $c > 0$ so that $\Omega_{r,c}$ is positively invariant.
- Consider the set $I = \{x \in \Omega_{r,c} \mid \omega = 0\}$.
- Again, which trajectories that start in I would remain in I ? We must have that if $\omega(0) = 0$ then $\dot{\omega}(t) = 0$ and $\omega(t) = 0$ for $t \geq 0$. Therefore, $-g\sin(\theta(t)) = 0 \Rightarrow \theta(t) = 0$ for $t \geq 0$. Hence, $M = \{(0,0)\}$ and the origin is asymptotically stable.

Example 3: Stability of pendulum

- However, unlike RLC circuit example, the point $(0,0)$ of pendulum not globally asymptotically stable.
- This is because pendulum also has equilibrium point at $(\pi,0)$. If one starts at this point and pendulum is not disturbed, it will stay there and not go to $(0,0)$.

Example 3: Stability of pendulum

- Now, we show that there is another choice of candidate Lyapunov function which is not physically motivated but immediately establishes asymptotic stability of $(0,0)$.
- Consider a new candidate Lyapunov function (keep in mind that $x = (\theta, \omega)$)

$$V(x) = \frac{1}{2}\omega^2 + \frac{1}{2}\left(\frac{1}{D}\omega + \theta\right)^2 + \frac{g}{r}\left(1 + \frac{1}{D^2}\right)(1 - \cos \theta)$$

Example 3: Stability of pendulum

- Now, by same calculations as before we have that

$$\dot{V}(x) = -(D\omega^2 + \frac{g}{rD}\theta \sin \theta)$$

- But, notice that if (θ, ω) is inside a disc of sufficiently small radius around 0 then $\theta \sin \theta \geq 0$ and is 0 in this disc only if $\theta = 0$. Thus

$$D\omega^2 + \frac{g}{rD}\theta \sin \theta = 0 \Rightarrow \omega^2 = 0, \theta \sin \theta = 0 \Rightarrow \omega = 0, \theta = 0.$$

Example 3: Stability of pendulum

- Inside this small disc we have that

$$\dot{V}(x) < 0 \text{ if } x = (\omega, \theta) \neq 0$$

- $V(x)$ lpd and $\dot{V}(x)$ lnd around equilibrium $(\omega, \theta) = (0, 0)$ (latter easy to check, tutorial exercise).
- Thus with new Lyapunov function we immediately establish asymptotic stability of the equilibrium point $(0, 0)$.